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$$\begin{aligned} & \int \frac{dx}{3x^2 + 2} \\ &= \frac{1}{2} \int \frac{dx}{\frac{3}{2}x^2 + 1} = \frac{1}{2} \cdot \frac{\sqrt{2}}{\sqrt{3}} \int \frac{d\left(\frac{\sqrt{3}}{\sqrt{2}}x\right)}{\left(\frac{\sqrt{3}}{\sqrt{2}}x\right)^2 + 1} \\ &= \frac{\sqrt{2}\sqrt{3}}{6} \operatorname{Bgtan} \frac{\sqrt{3}}{\sqrt{2}}x + c \\ &= \frac{\sqrt{6}}{6} \operatorname{Bgtan} \frac{\sqrt{6}}{2}x + c \end{aligned}$$

References